

JANNITTA YAO

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EDUCATION

Dartmouth College PhD Candidate in Earth Sciences	2022 - present
Wellesley College BA in Geosciences & Computer Science Durant Scholars Magna Cum Laude (GPA: 3.75/4.0)	2017 - 2021

FELLOWSHIPS, GRANTS, & AWARDS

National	
NSF Graduate Research Fellowship (\$159,000)	2024 - 2029
GSA Graduate Research Grant (\$2960)	2025
NSF AGeS3-Grad Geochronology Research Grant (\$9,588)	2025
GSA Robert D. Hatcher Research Award (\$2270)	2024
NEGSA Kenneth N. Weaver Student Travel Grant (\$150)	2022
GSA MGPV Travel Grant (\$500)	2022
Dartmouth College	
John A. Ebers 1961 Memorial Prize - Outstanding Graduate TA	2026
Irving Institute Graduate Energy Fellowship	2024 - 2025
Wellesley College	
Harold E. Andrews Award	2021
Sara F. Langer Memorial Award	2020

RESEARCH AND WORK EXPERIENCE

Post-Baccalaureate Research Assistant <i>Wellesley College Geosciences Department</i> Researched metamorphic and tectonic history of the Nashoba terrane in MA and of New York City with Professor Adrian Castro, assisted with department collections work, provided class and field trip support for multiple courses.	Sept 2021 - July 2022
Undergraduate Researcher <i>Castro Lab @ Wellesley College</i> Modelled effects of overstepped garnet nucleation on subduction zone devolatilization. Used a combination of petrography and thermodynamic modelling to constrain P-T conditions of metamorphic events in the Nashoba Terrane, MA.	Aug 2020 - July 2021
Peer Tutor <i>Wellesley College Geoscience and Computer Science Depts.</i> GEOS 101/102: The Dynamic Earth and CS 251: Theory of Programming Languages.	Aug 2020 - June 2021
Software Engineering Intern <i>Google</i> Trained Tensorflow ML model to predict throughput for Youtube's edge streaming servers.	May 2020 - Aug 2020
Research Assistant <i>MIT Darwin Project</i> Created and refined a Julia-based biogeochemical model of plankton in the global oceans.	Feb 2019 - Apr 2020
Engineering Practicum Intern <i>Google</i>	May 2019 - Aug 2019

Modernized Python-based dashboards used by software and site reliability engineers to monitor the status of Google Flights Search.

Teaching Assistant

June 2018 - Aug 2018

Girls Who Code

Taught the foundations of computer science including Scratch, Python, robotics, web development, and data science to female-identifying high school students.

Research Assistant

Jan 2018 - May 2018

Perron Lab @ MIT

Analyzed effect of wave energy and depth on coral reef spur and groove morphology in Molokai, Hawaii.

PUBLICATIONS

Yao, J., Keller, C.B., *in review*. A Bayesian Inversion for the Generation of Archean Felsic Crust. *G-cubed*.

Yao, J., Castro, A.E., Roberts, N.M. Wolfe, O.M., Brunet, I.M., Vasey, D.A., (2023). Phase Equilibria Evidence for Intermediate P/T Metamorphism in the Nashoba Terrane and Implications for Acadian Tectonics in Ganderia. *Lithosphere*.

CONFERENCE PRESENTATIONS

Geological Society of America Meeting

Oct 2026

New Constraints on Alleghanian (300 Ma) Amphibolite Facies Metamorphism in Central New England from Garnet Sm-Nd Petrochronology

Yao, J., Pansera, G., Castro, A.E., Brigham, J.M., Lee, A.S., Bista, D., Keller, C.B., Baxter, E.F., Strauss, J.V.

New Constraints on the Paleozoic Tectonic Evolution of Central New England from Metamorphic Petrochronology
Pansera, G., Castro, A.E., **Yao, J.**, Viete, D.R., Brenner, D.C.

Goldschmidt Conference

July 2026

A Bayesian Approach to the Generation of Archean Continental Crust (*talk*)

Yao, J., Keller, C.B.

Geological Society of America Meeting

Oct 2025

The Petrogenesis of the 1.7 Ga Twilight Trondhjemite in the Needle Mountains, CO

Yao, J., Keller, C.B.

American Geophysical Union Meeting

Dec 2024

A Bayesian Approach to the Generation of Archean Continental Crust (*talk*)

Yao, J., Keller, C.B.

American Geophysical Union Meeting

Dec 2023

Making Archean Crust: Modeling the Geochemical Evolution of Earth's Continents

Yao, J., Keller, C.B.

Gordon Geochronology Research Conference

Aug 2023

Making Archean Crust: Modeling the Geochemical Evolution of Earth's Continents

Yao, J., Keller, C.B.

American Geophysical Union Meeting

Dec 2022

Made in Manhattan: Insights into the Orogenic Evolution of the Manhattan Prong from Phase Equilibria and Geochronologic Constraints.

Castro, A.E., Tailby, N., **Yao, J.**, Brunet, I.M., Zizka, E.L., Van Scoy, E.

Geological Society of America Meeting

Oct 2022

"Barrovian" Metamorphism in the Nashoba Terrane: Evidence from Combined Phase Equilibria Modeling, Geothermobarometry, and Petrographic Analysis

Yao, J., Castro, A.E., Brunet, I.M., Roberts, N.M., Wolfe, O.M., Kuiper, Y.D., Hepburn, J.C.

Goldschmidt Conference

July 2022

Cryptic Early "Barrovian" Metamorphism in the Trailing Edge of Ganderia: Evidence from Aluminosilicate Reaction Textures, Garnet Trace Element Zoning, and Phase Equilibria Modeling

Castro, A.E., **Yao, J.**, Wolfe, O., George, F.R., Vasey, D.A.

NE Geological Society of America Meeting

Mar 2022

Synthesizing Petrographic Analysis and Modern Thermodynamic Modeling to Determine the P-T Conditions of Peak Metamorphism in the Nashoba Terrane, East-Central MA.

Yao, J., Brunet, I.M., Castro, A.E., Wolfe, O., Kuiper, Y., Hepburn, J.C.

Searching for the Suture: Applying Geochemical Techniques to Unravel the Nature of Cameron's Line in Central Park, NYC.

Zizka, E.L., Van Scoy, E., Brunet, I.M., Erwin, M.C., **Yao, J.**, Castro, A.E., Tailby, N.

NE Geological Society of America Meeting

Mar 2021

Garnet nucleation overstepping: A potential mechanism for fluid production within the seismic-aseismic transition in subduction zones.

Yao, J., Castro, A.E., Kohn, M.J., and Spear, F.S.

OUTREACH & COMMUNITY

EarthChem task force member

July 2026 - present

IEDA2

Advisory member for EarthChem task force on the future and maintenance of EarthChem post NSF-funding.

E.E. Just Graduate Fellow

Aug 2023 - present

Dartmouth College E.E. Just Program

Ensure success of underrepresented groups in STEM at Dartmouth and beyond by providing support through hosting a pre-orientation STEM workshop for incoming undergraduates, connecting students with research mentors, and providing resources and guidance for graduate school applications.

Volunteer Explainer and Guest Scientist

July 2023 - present

Montshire Museum of Science

Lead science and engineering activities for students in K-8 grade. Developed activities about geology for adult-only nights at the museum.

Like a Rock Committee

July 2023 - Oct 2024

Dartmouth Earth Sciences

Organized events to welcome incoming graduate students to the department.

Committee for Addressing Racism and Equity Representative

Oct 2022 - May 2023

Dartmouth College Graduate Student Council

Helped plan events for the graduate student community at Dartmouth to celebrate diversity and sent aid from graduate students to communities affected by Hurricane Ian in fall 2022.

Diversity, Equity, and Inclusion Committee

Sept 2021 - July 2022

Wellesley College Geosciences Department

Spearheaded DEI Committee by improving a diversity in geosciences project in the introductory course, updating the website with resources for current majors, and organizing professional development panels.

ADVISING & MENTORSHIP

Jaya L. Sharma (Dartmouth Senior Thesis Co-Advisor)
Geena San Diego (Dartmouth Early Research in the Sciences)
Lina Kim (Dartmouth Early Research in the Sciences)

SOFTWARE SKILLS

Programming	Julia, Python, Java, C++, MATLAB, Racket, C#
Software & Technology	Git/Github, Jupyter, Tensorflow, AWS, GIS, Unity, LaTeX, XMapTools, NSF ACCESS co-PI